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SOFTWARE REQUIREMENTS SPECIFICATION
Inward / Outward Document Management System

Document: IODMS/SRS/001
Version: 1.0 (Draft for Approval) | Standard: IEEE 830-1998
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Field	Value
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1.0	16/6/2026	Initial Draft, based on old software feature inventory	

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, and external interface requirements for the redevelopment of the Inward / Outward Document Management System (IODMS) at DEA, AURDC. Prepared per IEEE 830-1998, this document serves as the baseline requirements artifact under the V-Model SDLC adopted for this project.

1.2 Scope

The IODMS is an internal web-based application (with on demand desktop application versions) deployed on the HAL Nashik LAN. It manages creation, tracking, registration, and retrieval of official inward and outward correspondence. It replaces the legacy Microsoft Access-based desktop application and migrates all existing records and document files to a PostgreSQL-backed web architecture.

- Accessible only within HAL Nashik LAN, no internet connectivity (air-gapped).
- Developed strictly in Python as the primary programming language.
- Database: PostgreSQL (migrating from Microsoft Access).
- Approximately 5000+ existing PDFs/JPEGs (from 2006 onwards) to be migrated.
- Compliant with the Indian Official Secrets Act, 1923.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
IODMS	Inward / Outward Document Management System
HAL	Hindustan Aeronautics Limited
SRS	Software Requirements Specification
SDD	Software Design Description
FR	Functional Requirement
NFR	Non-Functional Requirement
EIR	External Interface Requirement
RTM	Requirements Traceability Matrix
RBAC	Role-Based Access Control
LAN	Local Area Network
OSA	Official Secrets Act, 1923
DOCX	Microsoft Word Document Format
Admin	System Administrator
TBD	To Be Determined: requires stakeholder confirmation before SDD

1.4 System Constraints

- **NETWORK:** Air-gapped LAN. No external internet access at any application layer.
- **COMPLIANCE:** Indian Official Secrets Act, 1923. No classified data outside HAL Nashik LAN.
- **LANGUAGE:** Python is the mandated primary development language.
- **DATABASE:** PostgreSQL. MS Access is inaccessible; legacy data available as 3 .xlsx exports only.

- USERS: ~30 registered users; peak concurrent: 10.
- FILE NUMBERING: Existing HAL numbering convention to be preserved. Format: TBD.
- STACK: Technology stack beyond Python and PostgreSQL is TBD - to be finalised before SDD.
- BROWSER: Target browsers TBD based on HAL workstation inventory.

1.5 References

- IEEE 830-1998: IEEE Recommended Practice for Software Requirements Specifications
- IEEE 1016-2009: IEEE Standard for Software Design Descriptions
- Indian Official Secrets Act, 1923
- HAL Nashik Legacy IODMS - UI Documentation (internal, 5 pages)
- Legacy Data Export - Inward Table, Outward Table, Address Book Table (.xlsx)

2. Functional Requirements

The following table captures all functional requirements derived from the legacy system and enhancement discussions. Priority levels: Critical > High > Medium > Low.

FR ID	Module	Requirement	Priority
FR-001	Create File	System shall provide a text input for entering the document subject.	High
FR-002	Create File	System shall provide a dropdown to select the creating officer from the registered user list.	High
FR-003	Create File	System shall provide a calendar date picker for the created-on date, defaulting to the current date.	High
FR-004	Create File	System shall provide a dropdown to select a file number; upon selection the corresponding file folder name shall be displayed automatically.	High
FR-005	Create File	System shall provide a dropdown to select an address category/group from the address book.	High
FR-006	Create File	System shall populate the 'Address To' list dynamically based on the selected address category.	High
FR-007	Create File	System shall display all selected primary addresses in a dedicated view area before submission.	High
FR-008	Create File	System shall allow the user to add CC recipients from the address book via a dropdown.	Medium
FR-009	Create File	System shall provide template selection for outward document type: Fax/Outside (with GM Signature), Fax/Outside (without GM Signature), or Internal Letter.	High
FR-010	Create File	System shall generate a pre-filled document based on the selected template and form data, which the user can review and edit before saving.	High
FR-011	Create File	System shall save the draft file to a temporary store (Finalise File queue) with a system-generated link address in the format fax-user-date-time.docx.	High
FR-012	Finalise File	System shall display all draft outward files in a table with columns: Link Address, Register No., Issuing Date, To, Subject, Remarks, Created By, Created On (date and time).	High
FR-013	Finalise File	System shall provide a 'Finalise File' action that auto-generates an Outward Register Number and records the date of finalisation.	High

FR ID	Module	Requirement	Priority
FR-014	Finalise File	System shall provide a 'Move to Trash' action for draft files.	Medium
FR-015	Finalise File	Once finalised, the file shall be moved from the draft queue and written to the Outward Register.	High
FR-016	Inward Entry	System shall auto-generate a Department Inward Number upon initiating a new inward entry.	High
FR-017	Inward Entry	System shall provide a calendar date picker for the date of inward, defaulting to the current date.	High
FR-018	Inward Entry	System shall provide a dropdown for scanned format selection and an option to trigger the connected scanner to scan a document directly.	High
FR-019	Inward Entry	System shall provide fields for Letter Reference Number, Letter Date, Received From (dropdown from address book), and Originated By (dropdown from address book).	High
FR-020	Inward Entry	System shall provide a text field for Subject.	High
FR-021	Inward Entry	System shall provide fields for Referred To, Folder Type (dropdown from address book), CC Sent To, and Remarks.	High
FR-022	Inward Entry	System shall allow classification of the inward entry as Type: Query / Snag / File.	High
FR-023	Inward Entry	System shall allow setting Status as Active (A) or Not Active (NA).	High
FR-024	Inward Entry	System shall support New, Modify, and Save operations for inward entries.	High
FR-025	Inward Register	System shall display all inward entries in a tabular register with columns: Receiving Date, Inward Number, Inward Letter No., Inward Date, Received From, Subject, Originated By, Referred To, File No., CC Sent To, Remarks, Type, Status.	High
FR-026	Inward Register	System shall provide a year-selection dropdown to filter the register view by year.	High
FR-027	Inward Register	System shall provide search functionality by: Referred To, Received From, Originated By, and Subject.	High
FR-028	Outward Register	System shall display all finalised outward entries in a tabular register with columns: Register No., File No., Issuing Date, To, Subject, Remarks, Created By.	High
FR-029	Outward Register	System shall provide a year-selection dropdown to filter the outward register by year.	High
FR-030	Outward Register	System shall provide search functionality by: File Number (dropdown), Created By (dropdown), Address To, and Subject.	High
FR-031	Address Book	System shall maintain an address book with fields: Address ID, Name, Designation, Organisation, Address Line 1, Address Line 2, Fax No., Email, Type.	High
FR-032	Address Book	System shall allow adding a new address entry and modifying an existing entry.	High
FR-033	Address Book	System shall auto-generate Address ID for new entries.	High
FR-034	Address Book	System shall provide search functionality by: Name, Designation, Organisation, Fax No., and Email using a dropdown selector and a text input.	High
FR-035	Address Book	Designation and Organisation fields shall be populated from managed dropdown lists (master lists maintained in Admin Panel).	Medium
FR-036	Address Book	System shall display address book data in a paginated table.	Medium

FR ID	Module	Requirement	Priority
FR-037	Admin Panel	System shall allow Admin to create new officer/user accounts with fields including name, designation (dropdown), department, and login credentials.	High
FR-038	Admin Panel	System shall allow Admin to activate, deactivate, or reset passwords for existing user accounts.	High
FR-039	Admin Panel	[PLACEHOLDER] Admin Panel shall include user management, audit log viewer, database backup trigger, yearly register reset, and master list management (designations, organisations, folder types). Detailed requirements TBD during design phase.	Medium
FR-040	Admin Panel	System shall enforce Role-Based Access Control (RBAC) with two roles: User and Admin. Admin has access to all modules; User is restricted to non-administrative modules.	High
FR-041	File Management	[TBD] System shall implement a file numbering scheme as defined by AURDC/DEA. File numbers shall reset yearly.	High
FR-042	File Management	System shall organise documents in a folder structure: I/O Register > Outward / Inward > Year > File (e.g., Su-30, LCA, etc.).	High
FR-043	Authentication	System shall provide login for User and Admin roles with username and password.	High
FR-044	Authentication	System shall provide a password reset mechanism for registered users.	High

3. Non-Functional Requirements

NFR ID	Category	Requirement	Priority
NFR-001	Security	The system shall operate exclusively on an air-gapped LAN network with no internet connectivity, in compliance with applicable GOI/HAL security directives including the Official Secrets Act 1923.	Critical
NFR-002	Security	All user passwords shall be stored in hashed form (bcrypt or equivalent). Plain-text password storage is prohibited.	Critical
NFR-003	Security	Session tokens shall expire after a configurable idle timeout (default: 30 minutes).	High
NFR-004	Security	All user actions on records (create, modify, finalise, delete) shall be logged with user ID, timestamp, and action type in an audit trail.	High
NFR-005	Performance	The system shall support up to 30 registered users and up to 10 concurrent users on the LAN without performance degradation.	High
NFR-006	Performance	Register table queries (inward/outward) shall return results within 3 seconds for datasets up to 50,000 records.	High
NFR-007	Reliability	The system shall achieve at least 99% uptime during office working hours on the server PC.	High
NFR-008	Reliability	Automated daily database backup shall be supported, triggered from Admin Panel or on schedule.	High
NFR-009	Usability	The UI shall be browser-based and compatible with current versions of Chromium-based browsers on Windows without any client-side installation.	High
NFR-010	Usability	All mandatory fields shall be visually marked; form validation errors shall be shown inline.	Medium
NFR-011	Maintainability	The backend shall be implemented in Python. Database shall be PostgreSQL. The system shall not depend on Microsoft Access.	Critical
NFR-012	Maintainability	Code shall be modular and documented to allow maintenance by a single developer or intern successor.	Medium
NFR-013	Scalability	The database schema shall support at least 10 years of document records (estimated 5,000+ documents per year) without schema changes.	Medium
NFR-014	Compatibility	The web application shall be packaged into a desktop application on demand (e.g., using Electron or equivalent) without changes to the core application code.	Medium
NFR-015	Data Integrity	Yearly register numbering reset shall be transactional; partial resets shall not corrupt existing records.	High
NFR-016	Data Migration	The system shall support one-time migration of existing data (~5,000+ records) from the legacy Microsoft Access database (available as XLSX) to PostgreSQL.	High

4. External Interface Requirements

EIR ID	Interface	Requirement	Priority
EIR-001	Web Browser (Client)	The application front-end shall render correctly on Chromium-based browsers (Chrome, Edge) without plugins or extensions.	High
EIR-002	Python Backend (Server)	The web server and API shall be implemented in Python. Framework TBD during design phase.	High
EIR-003	PostgreSQL Database	All application data shall be persisted in a PostgreSQL database hosted on the central server PC on the LAN.	Critical
EIR-004	Scanner Interface	The system shall provide an interface to trigger document scanning from the department's existing scanner. Scanner driver/SDK specifications TBD.	Medium
EIR-005	LAN Network	The system shall communicate over the internal LAN only. No external network calls shall be made by the application.	Critical
EIR-006	File System	Generated outward documents (.docx) shall be stored on the server file system in the defined folder hierarchy (I/O Register > Outward/Inward > Year > File).	High
EIR-007	Document Templates	The system shall read .docx templates from a designated template directory on the server and populate them with form data to produce draft outward documents.	High
EIR-008	Legacy XLSX Data	The system shall accept the three legacy Access-derived XLSX files as input for one-time data migration to PostgreSQL.	High
EIR-009	Desktop Packaging	The web application shall be packageable as a standalone desktop application (e.g., Electron) on demand, interfacing with the same backend API.	Medium
EIR-010	Audit Log Interface	The Admin Panel shall expose an interface to view system audit logs stored in the database.	Medium

5. Requirements Traceability Matrix

Columns 4 and 5 (Design Reference and Test Case Reference) are reserved for population during subsequent V-Model phases (High-Level Design and Test Planning).

Req ID	Requirement Title	Rationale / Notes	Design Ref (TBD)	Test Case Ref (TBD)
FR-001	Create File - System shall provide a text input for entering the docu...	Derived from legacy software feature.		
FR-002	Create File - System shall provide a dropdown to select the creating ...	Derived from legacy software feature.		
FR-003	Create File - System shall provide a calendar date picker for the cre...	Derived from legacy software feature.		
FR-004	Create File - System shall provide a dropdown to select a file number...	Derived from legacy software feature.		
FR-005	Create File - System shall provide a dropdown to select an address ca...	Derived from legacy software feature.		
FR-006	Create File - System shall populate the 'Address To' list dynamically...	Derived from legacy software feature.		
FR-007	Create File - System shall display all selected primary addresses in ...	Derived from legacy software feature.		
FR-008	Create File - System shall allow the user to add CC recipients from t...	Enhancement identified during requirements analysis.		
FR-009	Create File - System shall provide template selection for outward doc...	Derived from legacy software feature.		
FR-010	Create File - System shall generate a pre-filled document based on th...	Derived from legacy software feature.		
FR-011	Create File - System shall save the draft file to a temporary store (...)	Derived from legacy software feature.		
FR-012	Finalise File - System shall display all draft outward files in a table...	Derived from legacy software feature.		
FR-013	Finalise File - System shall provide a 'Finalise File' action that auto...	Derived from legacy software feature.		
FR-014	Finalise File - System shall provide a 'Move to Trash' action for draft...	Enhancement identified during requirements analysis.		
FR-015	Finalise File - Once finalised, the file shall be moved from the draft ...	Derived from legacy software feature.		
FR-016	Inward Entry - System shall auto-generate a Department Inward Number u...	Derived from legacy software feature.		
FR-017	Inward Entry - System shall provide a calendar date picker for the dat...	Derived from legacy software feature.		
FR-018	Inward Entry - System shall provide a dropdown for scanned format sele...	Derived from legacy software feature.		
FR-019	Inward Entry - System shall provide fields for Letter Reference Number...	Derived from legacy software feature.		

Req ID	Requirement Title	Rationale / Notes	Design Ref (TBD)	Test Case Ref (TBD)
FR-020	Inward Entry - System shall provide a text field for Subject.	Derived from legacy software feature.		
FR-021	Inward Entry - System shall provide fields for Referred To, Folder Typ...	Derived from legacy software feature.		
FR-022	Inward Entry - System shall allow classification of the inward entry a...	Derived from legacy software feature.		
FR-023	Inward Entry - System shall allow setting Status as Active (A) or Not ...	Derived from legacy software feature.		
FR-024	Inward Entry - System shall support New, Modify, and Save operations f...	Derived from legacy software feature.		
FR-025	Inward Register - System shall display all inward entries in a tabular re...	Derived from legacy software feature.		
FR-026	Inward Register - System shall provide a year-selection dropdown to filte...	Derived from legacy software feature.		
FR-027	Inward Register - System shall provide search functionality by: Referred ...	Derived from legacy software feature.		
FR-028	Outward Register - System shall display all finalised outward entries in a...	Derived from legacy software feature.		
FR-029	Outward Register - System shall provide a year-selection dropdown to filte...	Derived from legacy software feature.		
FR-030	Outward Register - System shall provide search functionality by: File Numb...	Derived from legacy software feature.		
FR-031	Address Book - System shall maintain an address book with fields: Addr...	Derived from legacy software feature.		
FR-032	Address Book - System shall allow adding a new address entry and modif...	Derived from legacy software feature.		
FR-033	Address Book - System shall auto-generate Address ID for new entries.	Derived from legacy software feature.		
FR-034	Address Book - System shall provide search functionality by: Name, Des...	Derived from legacy software feature.		
FR-035	Address Book - Designation and Organisation fields shall be populated ...	Enhancement identified during requirements analysis.		
FR-036	Address Book - System shall display address book data in a paginated t...	Enhancement identified during requirements analysis.		
FR-037	Admin Panel - System shall allow Admin to create new officer/user acc...	Derived from legacy software feature.		
FR-038	Admin Panel - System shall allow Admin to activate, deactivate, or re...	Derived from legacy software feature.		
FR-039	Admin Panel - [PLACEHOLDER] Admin Panel shall include user management...	Enhancement identified during requirements analysis.		

Req ID	Requirement Title	Rationale / Notes	Design Ref (TBD)	Test Case Ref (TBD)
FR-040	Admin Panel - System shall enforce Role-Based Access Control (RBAC) w...	Derived from legacy software feature.		
FR-041	File Management - [TBD] System shall implement a file numbering scheme as...	Derived from legacy software feature.		
FR-042	File Management - System shall organise documents in a folder structure: ...	Derived from legacy software feature.		
FR-043	Authentication - System shall provide login for User and Admin roles wit...	Derived from legacy software feature.		
FR-044	Authentication - System shall provide a password reset mechanism for reg...	Derived from legacy software feature.		
NFR-001	Security - The system shall operate exclusively on an air-gapped L...	Non-functional constraint for system quality.		
NFR-002	Security - All user passwords shall be stored in hashed form (bcry...	Non-functional constraint for system quality.		
NFR-003	Security - Session tokens shall expire after a configurable idle t...	Non-functional constraint for system quality.		
NFR-004	Security - All user actions on records (create, modify, finalise, ...	Non-functional constraint for system quality.		
NFR-005	Performance - The system shall support up to 30 registered users and ...	Non-functional constraint for system quality.		
NFR-006	Performance - Register table queries (inward/outward) shall return re...	Non-functional constraint for system quality.		
NFR-007	Reliability - The system shall achieve at least 99% uptime during off...	Non-functional constraint for system quality.		
NFR-008	Reliability - Automated daily database backup shall be supported, tri...	Non-functional constraint for system quality.		
NFR-009	Usability - The UI shall be browser-based and compatible with curre...	Non-functional constraint for system quality.		
NFR-010	Usability - All mandatory fields shall be visually marked; form val...	Non-functional constraint for system quality.		
NFR-011	Maintainability - The backend shall be implemented in Python. Database sh...	Non-functional constraint for system quality.		
NFR-012	Maintainability - Code shall be modular and documented to allow mainten...	Non-functional constraint for system quality.		
NFR-013	Scalability - The database schema shall support at least 10 years of ...	Non-functional constraint for system quality.		
NFR-014	Compatibility - The web application shall be packaged into a desktop ap...	Non-functional constraint for system quality.		
NFR-015	Data Integrity - Yearly register numbering reset shall be transactional;...	Non-functional constraint for system quality.		

Req ID	Requirement Title	Rationale / Notes	Design Ref (TBD)	Test Case Ref (TBD)
NFR-016	Data Migration - The system shall support one-time migration of existing...	Non-functional constraint for system quality.		
EIR-001	Web Browser (Client) - The application front-end shall render correctly on Chr...	Interface dependency identified from system context.		
EIR-002	Python Backend (Server) - The web server and API shall be implemented in Python. ...	Interface dependency identified from system context.		
EIR-003	PostgreSQL Database - All application data shall be persisted in a PostgreSQL...	Interface dependency identified from system context.		
EIR-004	Scanner Interface - The system shall provide an interface to trigger docume...	Interface dependency identified from system context.		
EIR-005	LAN Network - The system shall communicate over the internal LAN only...	Interface dependency identified from system context.		
EIR-006	File System - Generated outward documents (.docx) shall be stored on ...	Interface dependency identified from system context.		
EIR-007	Document Templates - The system shall read .docx templates from a designated...	Interface dependency identified from system context.		
EIR-008	Legacy XLSX Data - The system shall accept the three legacy Access-derived...	Interface dependency identified from system context.		
EIR-009	Desktop Packaging - The web application shall be packageable as a standalon...	Interface dependency identified from system context.		
EIR-010	Audit Log Interface - The Admin Panel shall expose an interface to view syste...	Interface dependency identified from system context.		